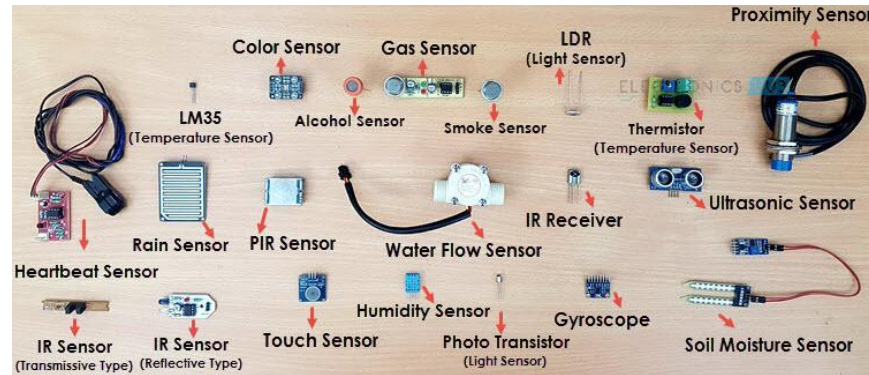


UNIT 3.SENSORS AND ACTUATORS IN IOT

3.1 DEFINITION OF SENSORS



A sensor is defined as a device that responds to a physical stimulus (such as heat, light, sound, pressure, magnetism, or a particular motion) and transmits a resulting impulse (as for measurement or operating a control).

A sensor is a device that detects changes in input stimuli/quantities like pressure, temperature, humidity, speed, etc, and generates an output quantity (mostly in the form of current or voltage) as a result of these changes.

3.2 TYPES OF SENSORS AND ITS USAGE

There are two types of sensors:

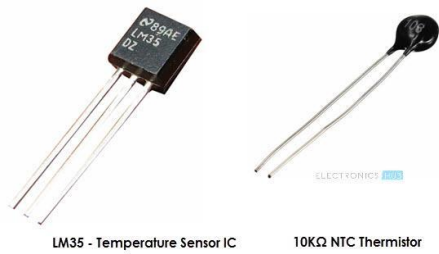
1. **Direct Sensor:** A sensor that can convert a non-electrical stimulus into an electrical signal with intermediate stages. Eg: thermocouple (temperature to voltage).
2. **Indirect Sensor:** A sensor that multiple conversion steps to transform the measured signal into an electrical signal. Eg: fiber-optic displacement sensor (light current to photons to current).

Now, remember all of the sensors can be classified mainly into these two types. But there are other ways to classify them.

1. Active and Passive Sensors
2. Contact and Non-contact sensors
3. Absolute and Relative Sensors

TEMPERATURE SENSORS:

Temperature is the commonly measured environmental quantity for various reasons. There are different types of temperature sensors that are used for measuring



temperature such as thermistors, resistance temperature detectors, thermocouples, semiconductor temperature sensors, and so on.

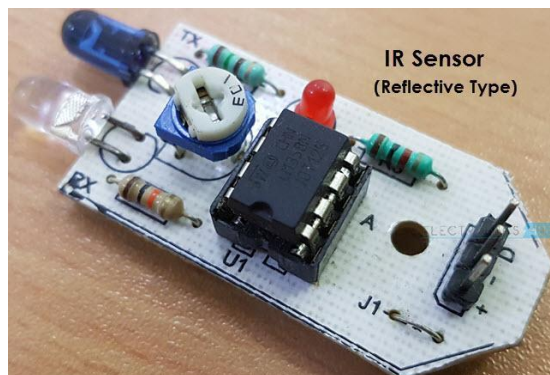
On the basis of requirements, various types of sensors are used for the purpose of measuring temperature in different applications. A simple temperature sensor with a circuit may be used for

switching the load on and off at a specific temperature.

This temperature is detected by the temperature sensor, in these cases, a thermistor is put to use. The temperature sensor circuit comprises a thermistor, transistor, relay, and battery. The temperature sensor activates the relay by detecting the required temperature.

The relay switches on the load that is connected to it which can be AC or DC. This circuit may be further utilized for the purpose of controlling the fan on the basis of temperature. Primarily, this type of sensor may be further classified into various other types such as digital temperature sensors, thermistors, and so on.

IR SENSORS:



The small photo chips comprising of photocells that are used for detecting and emitting infrared light are termed IR sensors. These sensors are most commonly used for the purpose of designing remote-control technology.

These sensors may be used for detecting various obstacles of the robotic vehicle and controlling the direction of the same. There are different types of sensors that might be

used for the detection of infrared lights.

A simple example of an IR sensor circuit that we use in our day-to-day life is a TV remote control. It comprises IR receiver circuits and IR emitter circuits that may be designed. The IR emitter circuit is used as a remote by the user for the purpose of emitting infrared light.

This infrared light is transmitted or sent to the IR receiver circuit which interfaces to the devices such as IR remote-controlled robots or a TV. These sensors are further used for designing television remote controls.

A TV remote is an example of a simple IR sensor-based electronics project which is used for the purpose of controlling a robotic vehicle in remote areas by using IR or TV remote. This type of TV remote is being utilized for sending commands to the robotic vehicle.

ULTRASONIC SENSOR:



An ultrasonic sensor or transceiver is a transducer (PRODUCE ELECTRICITY FROM OTHER ENERGY) that works on the principle like radar or sonar and is known for estimating the attributes of the target by interpreting. These sensors are classified as active and passive ultrasonic sensors and maybe differentiated on their working.

The active ultrasonic sensors are known for generating high-frequency sound waves that are received back by the ultrasonic sensor for the evaluation of echo. The time interval taken for receiving and transmitting the echo helps in the determination of the distance to an object.

However, the passive ultrasonic sensors are merely used for the detection of ultrasonic noise whose presence can be found in specific conditions. When it comes to the practical application of an ultrasonic sensor with a circuit, it may also be used as an ultrasonic distance sensor circuit.

LIGHT SENSOR

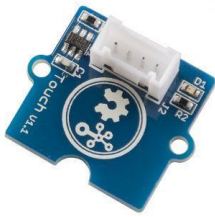


Sometimes also known as Photo Sensors, Light Sensors are one of the important sensors. A simple Light Sensor available today is the Light Dependent Resistor or LDR. The property of LDR is that its resistance is inversely proportional to the intensity of the ambient light i.e., when the intensity of light increases, its resistance decreases and vice-versa.

By using LDR in a circuit, we can calibrate the changes in its resistance to measure the intensity of Light. There are two other Light Sensors (or Photo Sensors) which are often used in complex electronic system design. They are Photo Diode and Photo Transistor. All these are Analog Sensors.

There are also Digital Light Sensors like BH1750, TSL2561, etc., which can calculate intensity of light and provide a digital equivalent value.

TOUCH SENSOR:



The switches that are activated by touch may be said to have touch sensors. These sensors are classified into different types on the basis of their touch-type such as piezo touch switch, capacitance touch switch, and resistance touch switch. For the purpose of controlling the load, a touch-sensitive load is designed. The touch-controlled load switch which is touch sensor principle-based comprises various blocks such as touch sensor plate, load, relay, and power supply block.

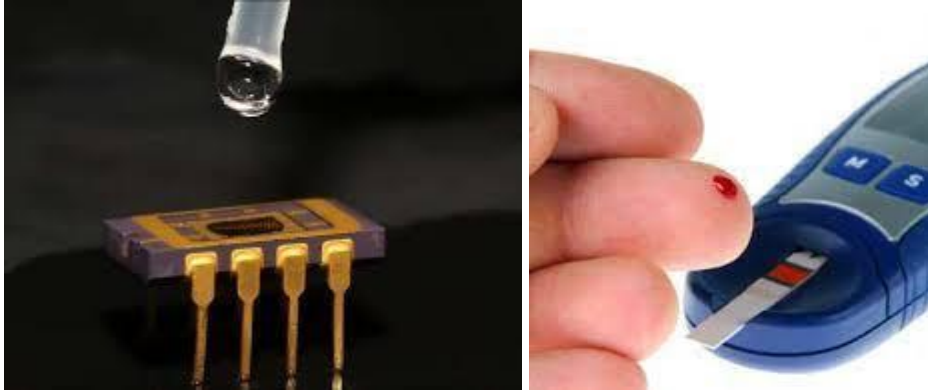
PROXIMITY SENSOR:



The proximity sensor is a type of IoT sensor in which the existence and non-existence of the surrounding objects are identified. After this, the detected signal is converted into a form that the user understands.

This type of sensor is mainly applied in the retail domain where any movement is found out and an association is present between the consumer and product. The users are provided with quick notifications related to exclusive offers and discount updates of the products in which they are interested.

CHEMICAL SENSOR:



A chemical sensor helps in determining various kinds of changes in liquid and for detecting the air chemical variations. These are mainly put to use in bigger cities and towns as it is important to look after the changes for providing safety to the people. This type of sensor is essentially implemented in commercially atmospheric observation. It is used for processing management which may comprise of chemicals, radioactive exposure, pharmaceutical industries, and reusable operations in space stations.

The most commonly used chemical sensors include chemical FET, electrochemical gas type, chemi resistor, non-dispersive IR, zinc oxide nanorod, and fluorescent chloride type.

GAS SENSOR:



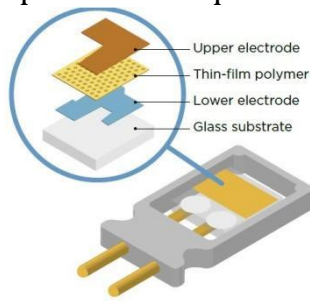
Gas sensors are similar to chemical sensors but are specially implemented for observing modifications in the quality of air. This is done to find out which types of gases are present in it. This sensor is being used in multiple domains such as health, manufacturing, agriculture, for supervision of gas in coal industries, chemical laboratory investigations, etc.

Some of the gas sensors that have been put to use include hydrogen type, hygrometer, carbon-dioxide sensor, ozone monitoring type, air pollution type, gas detection type, etc.

HUMIDITY SENSORS:

Humidity is used for defining the vapor present in the atmosphere or in gaseous substances. Humidity sensors are used as temperature sensors as the manufacturing

operations require similar operating conditions. By measuring humidity, it can be ensured that the complete procedure will pass away easily.



There are numerous domains in which these are used such as ventilators, residential and commercial use for heating purposes, etc. Some other domains include meteorology, greenhouses, automobile, painting, coating industries, and hospitals.

ACCELERATION SENSOR:



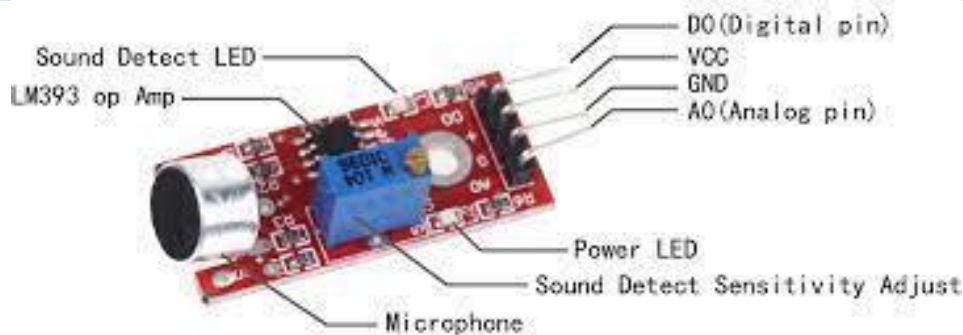
An acceleration sensor is used for calculating the acceleration and angular values. The accelerometer is primarily used for calculating acceleration. There are two types of forces that create an impact on the accelerometer; these are static force and dynamic force.

These sensors are present in numerous configurations and their type of selection is dependent on the industry's requirement. There are few parameters that are required to be checked before selecting the sensors such as total

number of axes, bandwidth, sensitivity, type of outputs, etc.

For example, accelerometers in laptops protect hard drives from damage. If the laptop were to suddenly drop while in use, the accelerometer would detect the sudden free fall and immediately turn off the hard drive to avoid hitting the reading heads into the hard drive platter.

SOUND SENSOR:



Sound sensors can generally be identified as the microphone devices that are used for delivering the corresponding level of voltage and sound on the basis of the sound level detection.

In comparison to light sensors, the design process of sound sensors is complicated. The reason being the delivery of minimal voltage difference by them. This is further to be amplified for providing voltage variation that is measurable.

3.3 INTRODUCTION TO ACTUATORS

One of the important components of interaction of mechanical, electrical and electronic engineering control system is the actuator. Actuators are the devices that accept a control command and produce a change in the physical system by generating force, motion, heat, flow, etc. It is a device that makes something to move.

Normally, the actuators are used in conjunction with the power supply and a coupling mechanism. The power unit provides either AC or DC power at the rated voltage and current. The coupling mechanism acts as the interface between the actuator and the physical system. Typical mechanisms include rack and pinion (wheel with teeth), gear drive, belt drive, lead screw and nut, piston, and linkages.

DEFINITION OF ACTUATORS :

- Actuators are the device used for converting hydraulic, pneumatic, and electrical energy into mechanical energy.
- The mechanical energy used to get the work done. Actuators perform functions just opposite to that of the pump.
- They can be used for lifting, tilting, clamping opening, closing, metering, mixing and turning, and for many other operations.

TYPES OF ACTUATORS :

1. ACTUATORS ARE BASICALLY CLASSIFIED INTO THE FOLLOWING TWO TYPES BASED ON THEIR MOVEMENT.

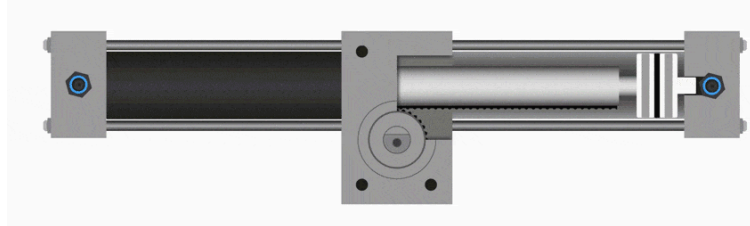
It is possible to convert one form of actuation to another. For example, translational motion is converted into rotational and vice versa by suitably designing the interfacing components around the actuator. Rack and pinion, gears, pulleys, lead screw and nut, and cams can be used as interfacing components.

- **LINEAR ACTUATORS:** Linear valve actuators are used to turn electric, pneumatic, and hydraulic energy into a push-and-pull motion or linear movement. A linear actuator can create both a forward and backward movement on a set linear plane. It is used in automation of food processing and packaging.



- **ROTARY ACTUATORS (ROTATIONAL):** Rotary actuators move in a circular motion. Machines that have rotating parts have a rotary actuator to complete a turning movement on a circular plane. Rotary actuators are extensively used on control valves and ball valves.

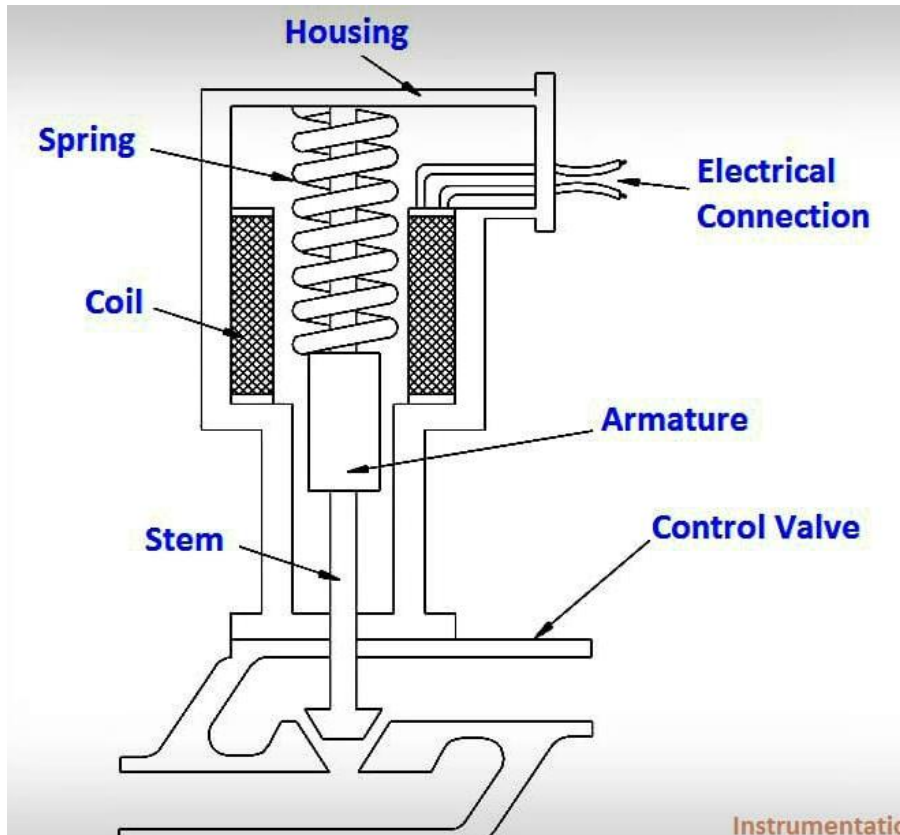
A rotary actuator can keep rotating in a given direction and they can go back and forth. Rotary valve actuators are often used in the oil and gas industry.



2. ACTUATORS CAN BE CLASSIFIED INTO THE FOLLOWING TYPES BASED ON THE TYPE OF ENERGY.

1. Electrical
2. Electromechanical
3. Electromagnetic
4. Hydraulic
5. Pneumatic

ELECTRICAL ACTUATORS :



An actuator obtaining electrical energy from the mechanical system is called electric actuators. Electrical actuators are equipped with a motor. The rotational energy of the motor is converted into the linear movement of a device.

This actuator possesses high positioning capabilities owing to electronic controllers fitted within. These capabilities along with velocity control are brilliantly exploited by various industries today.

These actuators are easy to set up, and can be reprogrammed quickly, and offer an immediate response. Electric actuators offer a wide range of motion profiles, owing to their versatile designs.

Electrical actuators comprise the following :

1. Drive system: DC motor, AC motor, Stepper motor

2. Switching Device:

a. Mechanical switch: Solenoids, Relays

b. Solid-state switch: Diodes, Thyristor, Transistors

ADVANTAGES OF ELECTRICAL ACTUATORS :

- High power conversion efficiency.
- The widespread availability of power supply.
- No pollution of the working environment.

- The basic drive element in an electric motor is usually lighter than that for fluid power.
- Structural components can be lightweight.
- The drive system is well suited to electronic control.

DISADVANTAGES OF ELECTRICAL ACTUATORS :

1. A larger and heavier motor must be used which is costly.
2. Poor dynamics response.
3. Compliance and wear problems are causing inaccuracies.
4. Conventional gear-driven creates a backlash.
5. Electric motors are not intrinsically safe. They cannot, therefore, be used in for explosive atmospheres.

ELECTRO-MECHANICAL ACTUATORS :

Electromechanical actuators are electrical motors that convert electrical energy into mechanical motion. Motors are the principal means of converting electrical energy into mechanical energy in industry. Broadly, they are classified into DC motors, AC motors, and stepper motors. It involves pulleys, chains, gears, rails, and other devices to operate. Example – Soft Actuators

Shape Memory Polymers

Light Activated Polymers

ELECTRO-MAGNETIC ACTUATORS

An electromagnetic actuator is a device that provides working motion due to an internal electromagnetic field. The solenoid is the most common electromagnetic actuator. A solenoid consists of a static, hollow-centered wire coil and a movable ferrous metal plunger. When the wire coil is energized, a magnetic field establishes that provides the force to push or pull the metal plunger.

HYDRAULIC AND PNEUMATIC ACTUATORS :

Hydraulic and pneumatic actuators are normally either rotary motors or linear pistons/cylinders. They are ideally suited for generating very large force coupled with large motion.

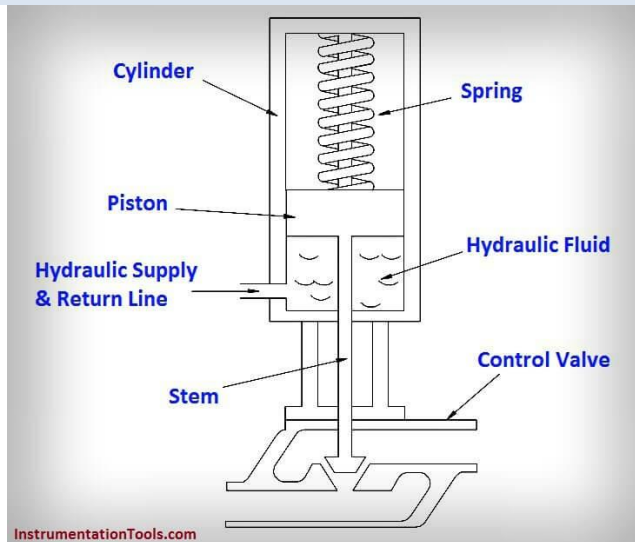
Pneumatic actuators use the compressed air that is more suitable for low to medium force, short stroke, and high-speed applications.

Hydraulic actuators use oil under pressure which is incompressible. They can produce very large force coupled with large motion in a cost-effective manner.

The rotary motors are usually used in applications where low speed and high torque are required. The cylinder/piston actuators are suited for the application of linear

motion. A proper selection of actuators and their drive systems for a particular application is of utmost importance in the design of mechatronic systems.

HYDRAULIC ACTUATORS :



- Hydraulic actuators transform the hydraulic energy stored in a reservoir into mechanical energy by means of suitable pumps.
- These actuators convert the energy of the fluid into mechanical energy and are capable of producing forces 25 times greater than pneumatic actuators. These actuators are differentiated into three types – hydraulic linear actuators, rotary actuators, and semi-rotary actuators – depending on the motion produced by them
- Hydraulic actuators are also fluid power device for industrial robots which utilize high-pressure fluid such as oil to transmit forces to the point of application desired.

ADVANTAGES OF HYDRAULIC ACTUATORS :

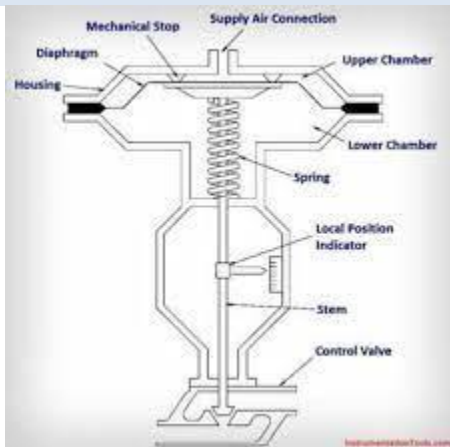
1. It has the advantage of generating extremely large force from a very compact actuator.
2. It can also provide precise control at low speeds
3. robust.
4. self-lubricating.
5. Due to the presence of an accumulator that acts as a storage device the system can meet sudden demand in power.
6. No mechanical linkages are required.
7. High efficiency and high power to size ratio.

DISADVANTAGES OF HYDRAULIC ACTUATORS :

1. The hydraulic system is required for a large infrastructure is high-pressure pump, tank, distribution lines.
2. Leakage can occur causing a loss in performance.
3. High maintenance.

4. Not suitable for a clean environment.

PNEUMATIC ACTUATORS



- Pneumatic actuators utilize pneumatic energy provided by a compressor and transforms it into mechanical energy by means of a piston or turbines.
- Pressurized air is used to transmit and control power.
- Pneumatic actuators are devices that cause things to move by taking advantage of potential energy.
- Pneumatic actuators are driven by air or gas, which is a clean operating medium. This medium helps prevent leakage or dripping, as in the case of many other actuators. Pneumatic actuators are known to convert compressed air into linear or rotary motion. The following advantages of these actuators have contributed to their popularity.

ADVANTAGES OF PNEUMATIC ACTUATORS :

The advantages of Pneumatic actuators are as follows,

1. Control is simple.
2. When the source of compressed air are readily available, as they often are in engineering related facilities, pneumatic actuators may be a good choice.
3. It is the cheapest form of all actuators
4. Pneumatic actuators have very quick action and response time, thus allowing for fast work cycles.
5. No mechanical transmission is usually required.

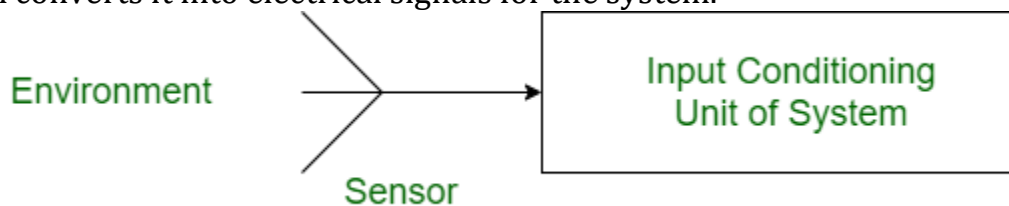
DISADVANTAGES OF PNEUMATIC ACTUATORS :

- More noise and vibration.
- Since air is compressible, pneumatic cylinders are not typically used for applications requiring accurate motion between two well-defined endpoints.
- Pneumatics are not suitable for heavy loads.
- If mechanical stops are used, resetting the system can be slow.

3.5 DIFFERENCE BETWEEN SENSORS & ACTUATORS

1. Sensor:

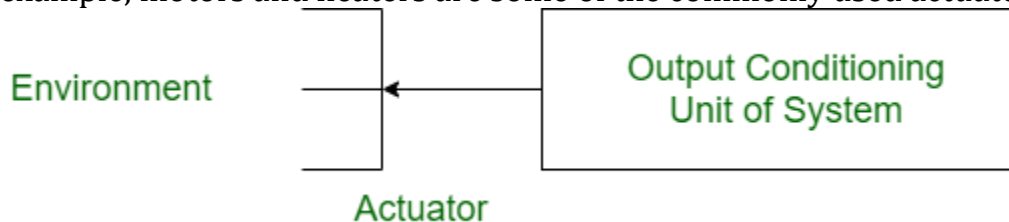
Sensor is a device used for the conversion of physical events or characteristics into the electrical signals. This is a hardware device that takes the input from environment and gives to the system by converting it. For example, a thermometer takes the temperature as physical characteristic and then converts it into electrical signals for the system.



2. Actuator:

Actuator is a device that converts the electrical signals into the physical events or characteristics. It takes the input from the system and gives output to the environment.

For example, motors and heaters are some of the commonly used actuators.



Difference between Sensor and Actuator :

SENSOR	ACTUATOR
It converts physical characteristics into electrical signals.	It converts electrical signals into physical characteristics.
It takes input from environment.	It takes input from output conditioning unit of system.
It gives output to input conditioning unit of system.	It gives output to environment.
Sensor generated electrical signals.	Actuator generates heat or motion.

SENSOR	ACTUATOR
It is placed at input port of the system.	It is placed at output port of the system.
It is used to measure the physical quantity.	It is used to measure the continuous and discrete process parameters.
It gives information to the system about environment.	It accepts command to perform a function.
Example: Photo-voltaic cell which converts light energy into electrical energy.	Example: Stepper motor where electrical energy drives the moto